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Electrical vapor provision system without wired electrical connection between a control unit and a cartridge

Abstract

An electronic vapor provision system includes a control unit and a cartridge configured to engage with and disengage from one another substantially along a longitudinal axis. The cartridge includes a reservoir of liquid to be vaporized, and the control unit includes a battery and a tube portion disposed along said longitudinal axis. The system further includes an induction heating assembly comprising a drive coil and a heater element. The heater element is located in the cartridge for vaporizing the liquid. The cartridge is configured to feed liquid from the reservoir onto the heater element for vaporization. The drive coil is located in the tube portion of the control unit and disposed about the longitudinal axis. The cartridge is located at least partly within the tube portion when engaged with the control unit, such that the heater element is within the drive coil.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation application of U.S. patent application Ser. No. 17/456,446 filed Nov. 24, 2021, which is a continuation of application Ser. No. 15/739,037 filed Dec. 21, 2017, which is a National Phase entry of PCT Application No. PCT/GB2016/051766, filed Jun. 15, 2016, which claims priority from GB Application No. 1511359.0, filed Jun. 29, 2015, all of which are hereby fully incorporated herein by reference.

FIELD

(1) The present disclosure relates to electronic vapor provision systems such as electronic nicotine delivery systems (e.g. e-cigarettes).

BACKGROUND

(2) FIG. 1 is a schematic diagram of one example of a conventional e-cigarette **10**. The e-cigarette has a generally cylindrical shape, extending along a longitudinal axis indicated by dashed line LA, and comprises two main components, namely a control unit **20** and a cartomizer **30**. The cartomizer **30** includes an internal chamber containing a reservoir of liquid formulation including nicotine, a vaporizer (such as a heater), and a mouthpiece **35**. The cartomizer **30** may further include a wick or similar facility to transport a small amount of liquid from the reservoir to the heater. The control unit **20** includes a re-chargeable battery to provide power to the e-cigarette **10** and a circuit board for generally controlling the e-cigarette **10**. When the heater receives power from the battery, as controlled by the circuit board, the heater vaporizes the nicotine and this vapor (aerosol) is then inhaled by a user through the mouthpiece **35**.

(3) The control unit **20** and cartomizer **30** are detachable from one another by separating in a direction parallel to the longitudinal axis LA, as shown in FIG. 1, but are joined together when the device **10** is in use by a connection, indicated schematically in FIG. 1 as **25A** and **25B**, to provide mechanical and electrical connectivity between the control unit **20** and the cartomizer **30**. The electrical connector on the control unit **20** that is used to connect to the cartomizer **30** also serves as a socket for connecting a charging device (not shown) when the control unit **20** is detached from the cartomizer **30**. The cartomizer **30** may be detached from the control unit **20** and disposed of when the supply of nicotine is exhausted (and replaced with another cartomizer if so desired).

(4) FIGS. 2 and 3 provide schematic diagrams of the control unit **20** and cartomizer **30** respectively of the e-cigarette **10** of FIG. 1. Note that various components and details, e.g. such as wiring and more complex shaping, have been omitted from FIGS. 2 and 3 for reasons of clarity. As shown in FIG. 2, the control unit **20** includes a battery or cell **210** for powering the e-cigarette **10**, as well as a chip, such as a (micro)controller for controlling the e-cigarette **10**. The controller **20** is attached to a small printed circuit board (PCB) **215** that also includes a sensor unit. If a user inhales on the mouthpiece **35**, air is drawn into the e-cigarette **10** through one or more air inlet holes (not shown in FIGS. 1 and 2). The sensor unit detects this airflow, and in response to such a detection, the controller provides power from the battery **210** to the heater in the cartomizer **30**.

(5) As shown in FIG. 3, the cartomizer **30** includes an air passage **161** extending along the central (longitudinal) axis LA of the cartomizer **30** from the mouthpiece **35** to the connector **25A** for joining the cartomizer **30** to the control unit **20**. A reservoir **170** of nicotine-containing liquid is provided around the air passage **161**. This reservoir **170** may be implemented, for example, by

providing cotton or foam soaked in the liquid. The cartomizer **30** also includes a heater **155** in the form of a coil for heating liquid from reservoir **170** to generate vapor to flow through air passage **161** and out through mouthpiece **35**. The heater **155** is powered through lines **166** and **167**, which are in turn connected to opposing polarities (positive and negative, or vice versa) of the battery **210** via connector **25A**.

(6) One end of the control unit provides a connector **25B** for joining the control unit **20** to the connector **25A** of the cartomizer **30**. The connectors **25A** and **25B** provide mechanical and electrical connectivity between the control unit **20** and the cartomizer **30**. The connector **25B** includes two electrical terminals, an outer contact **240** and an inner contact **250**, which are separated by insulator **260**. The connector **25A** likewise includes an inner electrode **175** and an outer electrode **171**, separated by insulator **172**. When the cartomizer **30** is connected to the control unit **20**, the inner electrode **175** and the outer electrode **171** of the cartomizer **30** engage the inner contact **250** and the outer contact **240**, respectively, of the control unit **20**. The inner contact **250** is mounted on a coil spring **255** so that the inner electrode **175** pushes against the inner contact **250** to compress the coil spring **255**, thereby helping to ensure good electrical contact when the cartomizer **30** is connected to the control unit **20**.

(7) The connector **25A** is provided with two lugs or tabs **180A**, **180B**, which extend in opposite directions away from the longitudinal axis LA of the e-cigarette **10**. These tabs **180A**, **180B** are used to provide a bayonet fitting for connecting the cartomizer **30** to the control unit **20**. It will be appreciated that other embodiments may use a different form of connection between the control unit **20** and the cartomizer **30**, such as a snap fit or a screw connection.

(8) As mentioned above, the cartomizer **30** is generally disposed of once the liquid reservoir **170** has been depleted, and a new cartomizer is purchased and installed. In contrast, the control unit **20** is re-usable with a succession of cartomizers **30**. Accordingly, it is particularly desirable to keep the cost of the cartomizer **30** relatively low. One approach to doing this has been to construct a three-part device, based on (i) a control unit, (ii) a vaporizer component, and (iii) a liquid reservoir. In this three-part device, only the final part, the liquid reservoir, is disposable, whereas the control unit and the vaporizer are both re-usable. However, having a three-part device can increase the complexity, both in terms of manufacture and user operation. Moreover, it can be difficult in such a three-part device to provide a wicking arrangement of the type shown in FIG. 3 to transport liquid from the reservoir to the heater.

(9) Another approach is to make the cartomizer **30** re-fillable, so that it is no longer disposable. However, making a cartomizer **30** re-fillable brings potential problems, for example, a user may try to re-fill the cartomizer **30** with an inappropriate liquid (one not provided by the supplier of the e-cigarette). There is a risk that this inappropriate liquid may result in a low quality consumer experience, and/or may be potentially hazardous, whether by causing damage to the e-cigarette itself, or possibly by creating toxic vapors.

(10) Accordingly, existing approaches for reducing the cost of a disposable component (or for avoiding the need for such a disposable component) have met with only limited success.

SUMMARY

(11) The invention is defined in the appended claims.

(12) Various embodiments of the disclosure provide an electronic vapor provision system having a longitudinal axis. The electronic vapor provision system comprises a control unit and a cartridge configured to engage with and disengage from one another substantially along said longitudinal axis. The cartridge includes a reservoir of liquid to be vaporized, and the control unit includes a battery and a tube portion disposed along said longitudinal axis. The system further comprises an induction heating assembly comprising a drive coil and a heater element. The heater element is located in the cartridge for vaporizing the liquid. The cartridge is configured to feed liquid from the reservoir onto the heater element for vaporization. The drive coil is located in the tube portion of the control unit and disposed about said longitudinal axis. The cartridge is located at least partly

within the tube portion when engaged with the control unit, such that the heater element is within the drive coil.

(13) The approach described herein is not restricted to specific embodiments such as set out below, but includes and contemplates any appropriate combinations of features presented herein. For example, an electronic vapor provision system may be provided in accordance with the approach described herein which includes any one or more of the various features described below as appropriate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various embodiments of the disclosure will now be described in detail by way of example only with reference to the following drawings:

(2) FIG. 1 is a schematic (exploded) diagram illustrating an example of a known e-cigarette.

(3) FIG. 2 is a schematic diagram of the control unit of the e-cigarette of FIG. 1.

(4) FIG. 3 is a schematic diagram of the cartomizer of the e-cigarette of FIG. 1.

(5) FIG. 4 is a schematic diagram illustrating an e-cigarette in accordance with some embodiments of the disclosure, showing the control unit assembled with the cartridge (top), the control unit by itself (middle), and the cartridge by itself (bottom).

(6) FIGS. 5 and 6 are schematic diagrams illustrating an e-cigarette in accordance with some other embodiments of the disclosure.

(7) FIG. 7 is a schematic diagram of the control electronics for an e-cigarette such as shown in FIGS. 4, 5 and 6 in accordance with some embodiments of the disclosure.

(8) FIGS. 7A, 7B and 7C are schematic diagrams of part of the control electronics for an e-cigarette such as shown in FIG. 6 in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

(9) FIG. 4 is a schematic diagram illustrating an e-cigarette **410** in accordance with some embodiments of the disclosure (please note that the term e-cigarette is used herein interchangeably with other similar terms, such as electronic vapor provision system, electronic aerosol provision system, etc.). The e-cigarette **410** includes a control unit **420** and a cartridge **430**. FIG. 4 shows the control unit **420** assembled with the cartridge **430** (top), the control unit **420** by itself (middle), and the cartridge **430** by itself (bottom). Note that for clarity, various implementation details (e.g. such as internal wiring, etc.) are omitted.

(10) As shown in FIG. 4, the e-cigarette **410** has a generally cylindrical shape with a central, longitudinal axis (denoted as LA, shown in dashed line). Note that the cross-section through the cylinder, i.e. in a plane perpendicular to the line LA, may be circular, elliptical, square, rectangular, hexagonal, or some other regular or irregular shape as desired. The general profile and shape (form factor) of the e-cigarette **410** is (or can be arranged to be) the same or broadly similar to that of e-cigarette **10** as shown in FIGS. 1-3. This consistency can be helpful for various reasons, such as potential to share components and packaging, user acceptance and ease of operation, brand recognition, etc.

(11) The mouthpiece **435** is located at one end of the cartridge **430**, while the opposite end of the e-cigarette **410** (with respect to the longitudinal axis) is denoted as the tip end **424**. The end of the cartridge **430** which is longitudinally opposite to the mouthpiece **435** is denoted by reference numeral **431**, while the end of the control unit **420** which is longitudinally opposite to the tip end **424** is denoted by reference numeral **421**.

(12) The cartridge **430** is able to engage with and disengage from the control unit **420** by movement along the longitudinal axis LA. More particularly, the end **431** of the cartridge **430** is able to engage with, and disengage from, the end **421** of the control unit **420**. Accordingly, ends

421 and **431** will be referred to as the engagement end **421** of the control unit **420** and the engagement end **431** of the cartridge **430**, respectively.

(13) The control unit **420** includes a battery **411** and a circuit board **415** to provide control functionality for the e-cigarette **410**, e.g. by provision of a controller, processor, Application Specific Integrated Circuit (ASIC) or similar form of control chip. The battery **411** is typically cylindrical in shape, and has a central axis that lies along, or at least close to, the longitudinal axis LA of the e-cigarette **410**. In FIG. 4, the circuit board **415** is shown longitudinally spaced from the battery **411**, in the opposite direction to the cartridge **430**. However, the skilled person will be aware of various other locations for the circuit board **415**, for example, it may be at the opposite end of the battery **411**. A further possibility is that the circuit board **415** lies along the side of the battery **411**—for example, with the e-cigarette **410** having a rectangular cross-section, the circuit board **415** located adjacent one outer wall of the e-cigarette **410**, and the battery **411** then slightly offset towards the opposite outer wall of the e-cigarette **410**. Note also that the functionality provided by the circuit board **415** (as described in more detail below) may be split across multiple circuit boards and/or across devices which are not mounted to a PCB, and these additional devices and/or PCBs can be located as appropriate within the e-cigarette **410**.

(14) The battery or cell **411** is generally re-chargeable, and one or more re-charging mechanisms may be supported. For example, a charging connection (not shown in FIG. 4) may be provided at the tip end **424**, and/or the engagement end **421** of the control unit **420**, and/or along the side of the e-cigarette **410**. Moreover, the e-cigarette **410** may support induction re-charging of battery **411**, in addition to (or instead of) re-charging via one or more re-charging connections or sockets.

(15) The control unit **420** includes a tube portion **440**, which extends along the longitudinal axis LA away from the engagement end **421** of the control unit **420**. The tube portion **440** is defined on the outside by outer wall **442**, which may generally be part of the overall outer wall or housing of the control unit **420**, and on the inside by inner wall **444**. A cavity **426** is formed by inner wall **444** of the tube portion and the engagement end **421** of the control unit **420**. This cavity **426** is able to receive and accommodate at least part of a cartridge **430** as it engages with the control unit **420** (as shown in the top drawing of FIG. 4).

(16) The inner wall **444** and the outer wall **442** of the tube portion define an annular space which is formed around the longitudinal axis LA. A drive coil **450** is located within this annular space, with the central axis of the drive coil **450** being substantially aligned with the longitudinal axis LA of the e-cigarette **410**. A drive coil **450** may sometimes also be referred to as a work coil. The drive coil **450** is electrically connected to the battery **411** and circuit board **415**, which provide power and control to the drive coil **450**, so that in operation, the drive coil **450** is able to provide induction heating to the cartridge **430**.

(17) The cartridge **430** includes a reservoir **470** containing liquid formulation (typically including nicotine). The reservoir **470** comprises a substantially annular region of the cartridge **430**, formed between an outer wall **476** of the cartridge **430**, and an inner tube or wall **472** of the cartridge **430**, both of which are substantially aligned with the longitudinal axis LA of the e-cigarette **410**. The liquid formulation may be held free within the reservoir **470**, or alternatively the reservoir **470** may be incorporated in some structure or material, e.g. sponge, to help retain the liquid within the reservoir **470**.

(18) The outer wall **476** has a portion **476A** of reduced cross-section of the cartridge **430**. This allows this portion **476A** of the reduced cross-section of the cartridge **430** to be received into the cavity **426** in the control unit **420** in order to engage the cartridge **430** with the control unit **420**. The remainder of the outer wall **476** has a greater cross-section in order to provide increased space within the reservoir **470**, and also to provide a continuous outer surface for the e-cigarette **410**—i.e. outer wall **476** of the cartridge is substantially flush with the outer wall **442** of the tube portion **440** of the control unit **420**. However, it will be appreciated that other implementations of the e-cigarette **410** may have a more complex/structured outer surface (compared with the smooth outer

surface shown in FIG. 4).

(19) The inside of the inner tube **472** defines a passageway **461** which extends, in a direction of airflow, from air inlet **461A** (located at the engagement end **431** of the cartridge **430** that engages the control unit **420**) through to air outlet **461B**, which is provided by the mouthpiece **435**. Located within the central passageway **461**, and hence within the airflow through the cartridge **430**, are heater element **455** and wick **454**. As can be seen in FIG. 4, the heater element **455** is located approximately in the center of the drive coil **450**. In particular, the location of the heater element **455** along the longitudinal axis LA can be controlled by having the step at the start of the portion **476A** of reduced cross-section for the cartridge **430** abut against the end (nearest the mouthpiece **435**) of the tube portion **440** of the control unit **420** (as shown in the top diagram of FIG. 4).

(20) The heater element **455** is made of a metallic material so as to permit use as a susceptor (or workpiece) in an induction heating assembly. More particularly, the induction heating assembly comprises the drive coil **450**, which produces a magnetic field having high frequency variations (when suitably powered and controlled by the battery **411** and controller on PCB **415**). This magnetic field is strongest in the center of the drive coil **450**, i.e. within cavity **426**, where the heater element **455** is located. The changing magnetic field induces eddy currents in the heater element **455** (which is capable of being conductively heated), thereby causing resistive heating within the heater element **455**. Note that the high frequency of the variations in magnetic field causes the eddy currents to be confined to the surface of the heater element **455** (via the skin effect), thereby increasing the effective resistance of the heater element **455**, and hence the resulting heating effect.

(21) Furthermore, the heater element **455** is generally selected to be a magnetic material having a high permeability, such as (ferrous) steel (rather than just a conductive material). In this case, the resistive losses due to eddy currents are supplemented by magnetic hysteresis losses (caused by repeated flipping of magnetic domains) to provide more efficient transfer of power from the drive coil **450** to the heater element **455**.

(22) The heater element **455** is at least partly surrounded by wick **454**. Wick **454** serves to transport liquid from the reservoir **470** onto the heater element **455** for vaporization. The wick **454** may be made of any suitable material, for example, a heat-resistant, fibrous material and typically extends from the passageway **461** through holes in the inner tube **472** to gain access into the reservoir **470**. The wick **454** is arranged to supply liquid to the heater element **455** in a controlled manner, in that the wick **454** prevents the liquid leaking freely from the reservoir **470** into passageway **461** (this liquid retention may also be assisted by having a suitable material within the reservoir **470** itself). Instead, the wick **454** retains the liquid within the reservoir **470**, and on the wick **454** itself, until the heater element **455** is activated, whereupon the liquid held by the wick **454** is vaporized into the airflow, and hence travels along passageway **461** for exit via mouthpiece **435**. The wick **454** then draws further liquid into itself from the reservoir **470**, and the process repeats with subsequent vaporizations (and inhalations) until the cartridge **430** is depleted.

(23) Although the wick **454** is shown in FIG. 4 as separate from (albeit encompassing) the heater element **455**, in some implementations, the heater element **455** and wick **454** may be combined together into a single component, such as a heater element **455** made of a porous, fibrous steel material which can also act as a wick **454** (as well as a heater). In addition, although the wick **454** is shown in FIG. 4 as supporting the heater element **455**, in other embodiments, the heater element **455** may be provided with separate supports, for example, by being mounted to the inner tube **472** (instead of or in addition to being supported by the heater element **455**).

(24) The heater element **455** may be substantially planar, and perpendicular to the central axis of the drive coil **450** and the longitudinal axis LA of the e-cigarette **410**, since induction primarily occurs in this plane. Although FIG. 4 shows the heater element **455** and wick **454** extending across the full diameter of the inner tube **472**, typically the heater element **455** and wick **454** will not cover the whole cross-section of the air passage-way **461**. Instead, space is typically provided to

allow air to flow through the inner tube **472** from air inlet **461A** and around heater element **455** and wick **454** to pick up the vapor produced by the heater. For example, when viewed along the longitudinal axis **LA**, the heater element **455** and wick **454** may have an “O” configuration with a central hole (not shown in FIG. **4**) to allow for airflow along the passageway **461**. Many other configurations are possible, such as the heater having a “Y” or “X” configuration. (Note that in such implementations, the arms of the “Y” or “X” would be relatively broad to provide better induction.)

(25) Although FIG. **4** shows the engagement end **431** of the cartridge **430** as covering the air inlet **461A**, this end of the cartomizer may be provided with one or more holes (not shown in FIG. **4**) to allow the desired air intake to be drawn into passageway **461**. Note also that in the configuration shown in FIG. **4**, there is a slight gap **422** between the engagement end **431** of the cartridge **430** and the corresponding engagement end **421** of the control unit **420**. Air can be drawn from this gap **422** through air inlet **461A**.

(26) The e-cigarette **410** may provide one or more routes to allow air to initially enter the gap **422**. For example, there may be sufficient spacing between the outer wall **476** of the cartridge **430** and the inner wall **444** of tube portion **440** to allow air to travel into gap **422**. Such spacing may arise naturally if the cartridge **430** is not a tight fit into the cavity **426**. Alternatively one or more air channels may be provided as slight grooves along one or both of these walls **476**, **444** to support this airflow. Another possibility is for the housing of the control unit **420** to be provided with one or more holes, firstly to allow air to be drawn into the control unit **420**, and then to pass from the control unit **420** into gap **422**. For example, the holes for air intake into the control unit **420** might be positioned as indicated in FIG. **4** by arrows **428A** and **428B**, and engagement end **421** of control unit **420** might be provided with one or more holes (not shown in FIG. **4**) for the air to pass out from the control unit **420** into gap **422** (and from there into the cartridge **430**). In other implementations, gap **422** may be omitted, and the airflow may, for example, pass directly from the control unit **420** through the air inlet **461A** into the cartridge **430**.

(27) The e-cigarette **410** may be provided with one or more activation mechanisms for the induction heater assembly, i.e. to trigger operation of the drive coil **450** to heat the heater element **455**. One possible activation mechanism is to provide a button **429** on the control unit, which a user may press to active the heater element **455**. This button may be a mechanical device, a touch sensitive pad, a sliding control, etc. The heater element **455** may stay activated for as long as the user continues to press or otherwise positively actuate the button **429**, subject to a maximum activation time appropriate to a single puff of the e-cigarette (typically a few seconds). If this maximum activation time is reached, the controller may automatically de-activate the induction heater to prevent over-heating. The controller may also enforce a minimum interval (again, typically for a few seconds) between successive activations.

(28) The induction heater assembly may also be activated by airflow caused by a user inhalation. In particular, the control unit **420** may be provided with an airflow sensor for detecting an airflow (or pressure drop) caused by an inhalation. The airflow sensor is then able to notify the controller of this detection, and the induction heater is activated accordingly. The induction heater may remain activated for as long as the airflow continues to be detected, subject again to a maximum activation time as above (and typically also a minimum interval between puffs).

(29) Airflow actuation of the heater element **455** may be used instead of providing button **429** (which could therefore be omitted), or alternatively the e-cigarette **410** may require dual activation in order to operate—i.e. both the detection of airflow and the pressing of button **429**. This requirement for dual activation can help to provide a safeguard against unintended activation of the e-cigarette **410**.

(30) It will be appreciated that the use of an airflow sensor generally involves an airflow passing through the control unit **420** upon inhalation, which is amenable to detection (even if this airflow only provides part of the airflow that the user ultimately inhales). If no such airflow passes through

the control unit **420** upon inhalation, then button **429** may be used for activation, although it might also be possible to provide an airflow sensor to detect an airflow passing across a surface of (rather than through) the control unit **420**.

(31) There are various ways in which the cartridge **430** may be retained within the control unit **420**. For example, the inner wall **444** of the tube portion **440** of the control unit **420** and the outer wall **476** of the portion **476A** of reduced cross-section of the cartridge **430** may each be provided with a screw thread (not shown in FIG. 4) for mutual engagement. Other forms of mechanical engagement, such as a snap fit, a latching mechanism (perhaps with a release button or similar) may also be used. Furthermore, the control unit **420** may be provided with additional components to provide a fastening mechanism, such as described below.

(32) In general terms, the attachment of the cartridge **430** to the control unit **420** for the e-cigarette **410** of FIG. 4 is simpler than in the case of the e-cigarette **10** shown in FIGS. 1-3. In particular, the use of induction heating for e-cigarette **410** allows the connection between the cartridge **430** and the control unit **420** to be mechanical only, rather than also having to provide an electrical connection with wiring to a resistive heater. Consequently, the mechanical connection may be implemented, if so desired, by using an appropriate plastic molding for the housing of the cartridge **430** and the control unit **420**; in contrast, in the e-cigarette **10** of FIGS. 1-3, the housings of the cartomizer **30** and the control unit **20** have to be somehow bonded to a metal connector. Furthermore, the connector of the e-cigarette **10** of FIGS. 1-3 has to be made in a relatively precise manner to ensure a reliable, low contact resistance, electrical connection between the control unit **20** and the cartomizer **30**. In contrast, the manufacturing tolerances for the purely mechanical connection between the cartridge **430** and the control unit **420** of e-cigarette **410** are generally greater. These factors all help to simplify the production of the cartridge **430** and thereby to reduce the cost of this disposable (consumable) component.

(33) Furthermore, conventional resistive heating often utilizes a metallic heating coil surrounding a fibrous wick, however, it is relatively difficult to automate the manufacture of such a structure. In contrast, an inductive heater element **455** is typically based on some form of metallic disk (or other substantially planar component), which is an easier structure to integrate into an automated manufacturing process. This again helps to reduce the cost of production for the disposable cartridge **430**.

(34) Another benefit of inductive heating is that conventional e-cigarettes may use solder to bond power supply wires to a resistive heater coil. However, there is some concern that heat from the coil during operation of such an e-cigarette might volatilize undesirable components from the solder, which would then be inhaled by a user. In contrast, there are no wires to bond to the heater element **455** (which is capable of being inductively heated), and hence the use of solder can be avoided within the cartridge **430**. Also, a resistive heater coil as in a conventional e-cigarette generally comprises a wire of relatively small diameter (to increase the resistance and hence the heating effect). However, such a thin wire is relatively delicate and so may be susceptible to damage, whether through some mechanical mistreatment and/or potentially by local overheating and then melting. In contrast, a disk-shaped heater element **455** as used for induction heating is generally more robust against such damage.

(35) FIGS. 5 and 6 are schematic diagrams illustrating an e-cigarette in accordance with some other embodiments of the disclosure. To avoid repetition, aspects of FIGS. 5 and 6 that are generally the same as shown in FIG. 4 will not be described again, except where relevant to explain the particular features of FIGS. 5 and 6. Note also that reference numbers having the same last two digits typically denote the same or similar (or otherwise corresponding) components across FIGS. 4 to 6 (with the first digit in the reference number corresponding to the Figure containing that reference number).

(36) In the e-cigarette **510** shown in FIG. 5, the control unit **520** is broadly similar to the control unit **420** shown in FIG. 4, however, the internal structure of the cartridge **530** is somewhat different

from the internal structure of the cartridge **430** shown in FIG. 4. Thus rather than having a central airflow passage, as for e-cigarette **410** of FIG. 4, in which the liquid reservoir **470** surrounds the central airflow passage **461**, in the e-cigarette **510** of FIG. 5, the air passageway **561** is offset from the central, longitudinal axis (LA) of the cartridge **530**. In particular, the cartridge **530** contains an inner wall **572** that separates the internal space of the cartridge **530** into two portions. A first portion, defined by inner wall **572** and one part of outer wall **576**, provides a chamber for holding the reservoir **570** of liquid formulation. A second portion, defined by inner wall **572** and an opposing part of outer wall **576**, defines the air passage way **561** through the e-cigarette **510**.

(37) In addition, the e-cigarette **510** does not have a wick, but rather relies upon a porous heater element **555** to act both as the heating element (susceptor) and the wick to control the flow of liquid out of the reservoir **570**. The porous heater element **555** may be made, for example, of a material formed from sintering or otherwise bonding together steel fibers.

(38) The heater element **555** is located at the end of the reservoir **570** opposite to the mouthpiece **535** of the cartridge **530**, and may form some or all of the wall of the reservoir chamber at this end. One face of the heater element is in contact with the liquid in the reservoir **570**, while the opposite face of the heater element **555** is exposed to an airflow region **538** which can be considered as part of air passageway **561**. In particular, this airflow region **538** is located between the heater element **555** and the engagement end **531** of the cartridge **530**.

(39) When a user inhales on mouthpiece **535**, air is drawn into the region **538** through the engagement end **531** of the cartridge **530** from gap **522** (in a similar manner to that described for the e-cigarette **410** of FIG. 4). In response to the airflow (and/or in response to the user pressing button **529**), the coil **550** is activated to supply power to heater **555**, which therefore produces a vapor from the liquid in reservoir **570**. This vapor is then drawn into the airflow caused by the inhalation, and travels along the air passageway **561** (as indicated by the arrows) and out through mouthpiece **535**.

(40) In the e-cigarette **610** shown in FIG. 6, the control unit **620** is broadly similar to the control unit **420** shown in FIG. 4, but now accommodates two (smaller) cartridges **630A**, and **630B**. Each of these cartridges **630A**, **630B** is analogous in structure to the portion **476A** of reduced cross-section of the cartridge **430** in FIG. 4. However, the longitudinal extent of each of the cartridges **630A** and **630B** is only half that of the portion **476A** of reduced cross-section of the cartridge **430** in FIG. 4, thereby allowing two cartridges to be contained within the region in e-cigarette **610** corresponding to cavity **426** in e-cigarette **410**, as shown in FIG. 4. In addition, the engagement end **621** of the control unit **620** may be provided, for example, with one or more struts or tabs (not shown in FIG. 6) that maintain cartridges **630A**, **630B** in the position shown in FIG. 6 (rather than closing the gap **622**).

(41) In the e-cigarette **610**, the mouthpiece **635** may be regarded as part of the control unit **620**. In particular, the mouthpiece **635** may be provided as a removable cap or lid, which can screw or clip onto and off the remainder of the control unit **620** (or any other appropriate fastening mechanism can be used). The mouthpiece **635** is removed from the rest of the control unit **620** to insert a new cartridge or to remove an old cartridge, and then fixed back onto the control unit for use of the e-cigarette **610**.

(42) The operation of the individual cartridges **630A**, **630B** in e-cigarette **610** is similar to the operation of cartridge **430** in e-cigarette **410**, in that each cartridge includes a wick **654A**, **654B** extending into the respective reservoir **670A**, **670B**. In addition, each cartridge **630A**, **630B** includes a heating element, **655A**, **655B**, accommodated in a respective wick, **654A**, **654B**, and may be energized by a respective coil **650A**, **650B** provided in the control unit **620**. The heaters **655A**, **655B** vaporize liquid into a common passageway **661** that passes through both cartridges **630A**, **630B** and out through mouthpiece **635**.

(43) The different cartridges **630A**, **630B** may be used, for example, to provide different flavors for the e-cigarette **610**. In addition, although the e-cigarette **610** is shown as accommodating two

cartridges **630A**, **630B**, it will be appreciated that some devices may accommodate a larger number of cartridges. Furthermore, although cartridges **630A** and **630B** are the same size as one another, some devices may accommodate cartridges of differing size. For example, an e-cigarette may accommodate one larger cartridge having a nicotine-based liquid, and one or more small cartridges to provide flavor or other additives as desired.

(44) In some cases, the e-cigarette **610** may be able to accommodate (and operate with) a variable number of cartridges. For example, there may be a spring or other resilient device mounted on control unit engagement end **621**, which tries to extend along the longitudinal axis towards the mouthpiece **635**. If one of the cartridges **630A**, **630B** shown in FIG. **6** is removed, this spring would therefore help to ensure that the remaining cartridge(s) would be held firmly against the mouthpiece **635** for reliable operation.

(45) If an e-cigarette has multiple cartridges, one option is that these are all activated by a single coil that spans the longitudinal extent of all the cartridges. Alternatively, there may be an individual coil **650A**, **650B** for each respective cartridge **630A**, **630B**, as illustrated in FIG. **6**. A further possibility is that different portions of a single coil may be selectively energized to mimic (emulate) the presence of multiple coils.

(46) If an e-cigarette does have multiple coils for respective cartridges (whether really separate coils, or emulated by different sections of a single larger coil), then activation of the e-cigarette (such as by detecting airflow from an inhalation and/or by a user pressing a button) may energize all coils. The e-cigarettes **410**, **510**, **610**, however, support selective activation of the multiple coils, whereby a user can choose or specify which coil(s) to activate. For example, e-cigarette **610** may have a mode or user setting in which in response to an activation, only coil **650A** is energized, but not coil **650B**. This would then produce a vapor based on the liquid formulation in coil **650A**, but not coil **650B**. This would allow a user greater flexibility in the operation of e-cigarette **610**, in terms of the vapor provided for any given inhalation (but without a user having to physically remove or insert different cartridges just for that particular inhalation).

(47) It will be appreciated that the various implementations of e-cigarette **410**, **510** and **610** shown in FIGS. **4-6** are provided as examples only, and are not intended to be exhaustive. For example, the cartridge design shown in FIG. **5** might be incorporated into a multiple cartridge device such as shown in FIG. **6**. The skilled person will be aware of many other variations that can be achieved, for example, by mixing and matching different features from different implementations, and more generally by adding, replacing and/or removing features as appropriate.

(48) FIG. **7** is a schematic diagram of the main electronic components of the e-cigarettes **410**, **510**, **610** of FIGS. **4-6** in accordance with some embodiments of the disclosure. The heater element **455**, which is located in the cartridge **430**, may comprise any suitable structure, or combination of structures, for inductive heating. The remaining elements shown in FIG. **7** are located in the control unit **420**. It will be appreciated that since the control unit **420** is a re-usable device (in contrast to the cartridge **430** which is a disposable or consumable), it is acceptable to incur one-off costs in relation to production of the control unit **420** which would not be acceptable as repeat costs in relation to the production of the cartridge **430**. The components of the control unit **420** may be mounted on circuit board **415**, or may be separately accommodated in the control unit **420** to operate in conjunction with the circuit board **415** (if provided), but without being physically mounted on the circuit board **415** itself.

(49) As shown in FIG. **7**, the control unit **420** includes a re-chargeable battery **411**, which is linked to a re-charge connector or socket **725**, such as a micro-USB interface. This connector **725** supports re-charging of battery **411**. Alternatively, or additionally, the control unit **420** may also support re-charging of battery **411** by a wireless connection (such as by induction charging).

(50) The control unit **420** further includes a controller **715** (such as a processor or application specific integrated circuit, ASIC), which is linked to a pressure or airflow sensor **716**. The controller **715** may activate the induction heating, as discussed in more detail below, in response to

the sensor **716** detecting an airflow. In addition, the control unit **420** further includes a button **429**, which may also be used to activate the induction heating, as described above.

(51) FIG. 7 also shows a comms/user interface **718** for the e-cigarette. This may comprise one or more facilities according to the particular implementation. For example, the comms/user interface **718** may include one or more lights and/or a speaker to provide output to the user, for example to indicate a malfunction, battery charge status, etc. The comms/user interface **718** may also support wireless communications, such as Bluetooth or near field communications (NFC), with an external device, such as a smartphone, laptop, computer, notebook, tablet, etc. The e-cigarette may utilize this comms/user interface **718** to output information such as device status, usage statistics, etc., to the external device, for ready access by a user. The comms/user interface **718** may also be utilized to allow the e-cigarette to receive instructions, such as configuration settings entered by the user into the external device. For example, the comms/user interface **718** and controller **715** may be utilized to instruct the e-cigarette to selectively activate different coils **650A**, **650B** (or portions thereof), as described above. In some cases, the comms/user interface **718** may use the drive coil **450** to act as an antenna for wireless communications.

(52) The controller **715** may be implemented using one or more chips as appropriate. The operations of the controller **715** are generally controlled at least in part by software programs running on the controller **715**. Such software programs may be stored in non-volatile memory, such as Read Only Memory (ROM), which can be integrated into the controller **715** itself, or provided as a separate component (not shown). The controller **715** may access the ROM to load and execute individual software programs as and when required.

(53) The controller **715** controls the inductive heating of the e-cigarette by determining when the device is or is not properly activated—for example, whether an inhalation has been detected, and whether the maximum time period for an inhalation has not yet been exceeded. If the controller **715** determines that the e-cigarette is to be activated for vaping, the controller **715** arranges for the battery **411** to supply power to the inverter **712**. The inverter **712** is configured to convert the DC output from the battery **411** into an alternating current signal, typically of relatively high frequency—e.g. 1 MHz (although other frequencies, such as 5 kHz, 20 kHz, 80 KHz, or 300 kHz, or any range defined by two such values, may be used instead). This Alternating Current (AC) signal is then passed from the inverter to the drive coil **450**, via suitable impedance matching (not shown in FIG. 7) if so required.

(54) The drive coil **450** may be integrated into some form of resonant circuit, such as by combining in parallel with a capacitor (not shown in FIG. 7), with the output of the inverter **712** tuned to the resonant frequency of this resonant circuit. This resonance causes a relatively high current to be generated in drive coil **450**, which in turn produces a relatively high magnetic field in heater element **455**, thereby causing rapid and effective heating of the heater element **455** to produce the desired vapor or aerosol output.

(55) FIG. 7A illustrates part of the control electronics for an e-cigarette **610** having multiple coils in accordance with some implementations (while omitting for clarity aspects of the control electronics not directly related to the multiple coils). FIG. 7A shows a power source **782A** (typically corresponding to the battery **411** and inverter **712** of FIG. 7), a switch configuration **781A**, and the two work coils **650A**, **650B**, each associated with a respective heater element **655A**, **655B** as shown in FIG. 6 (but not included in FIG. 7A). The switch configuration has three outputs denoted A, B and C in FIG. 7A. It is also assumed that there is a current path between the two work coils **650A**, **650B**.

(56) In order to operate the induction heating assembly, two out of three of these outputs A, B, C are closed (to permit current flow), while the remaining output stays open (to prevent current flow). Closing outputs A and C activates both coils, and hence both heater elements **655A**, **655B**; closing A and B selectively activates just work coil **650A**; and closing B and C activates just work coil **650B**.

(57) Although it is possible to treat work coils **650A** and **650B** just as a single overall coil (which is either on or off together), the ability to selectively energize either or both of work coils **650A** and **650B**, such as provided by the implementation of FIG. 7, has a number of advantages, including: a) choosing the vapor components (e.g. flavorants) for a given puff. Thus activating just work coil **650A** produces vapor just from reservoir **670A**; activating just work coil **650B** produces vapor just from reservoir **670B**; and activating both work coils **650A**, **650B** produces a combination of vapor from both reservoirs **670A**, **670B**. b) controlling the amount of vapor for a given puff. For example, if reservoir **670A** and reservoir **670B** in fact contain the same liquid, then activating both work coils **650A**, **650B** can be used to produce a stronger (higher vapor level) puff compared to activating just one work coil by itself c) prolonging battery (charge) lifetime. As already discussed, it may be possible to operate the e-cigarette **610** of FIG. 6 when it contains just a single cartridge, e.g. **630B** (rather than also including cartridge **630A**). In this case, it is more efficient just to energize the work coil **650B** corresponding to cartridge **630B**, which is then used to vaporize liquid from reservoir **670B**. In contrast, if the work coil **650A** corresponding to the (missing) cartridge **630A** is not energized (because this cartridge and the associated heater element **650A** are missing from e-cigarette **610**), then this saves power consumption without reducing vapor output.

(58) Although the e-cigarette **610** of FIG. 6 has a separate heater element **655A**, **655B** for each respective work coil **650A**, **650B**, in some implementations, different work coils may energize different portions of a single (larger) workpiece or susceptor. Accordingly, in such an e-cigarette **610**, the different heater elements **655A**, **655B** may represent different portions of the larger susceptor, which is shared across different work coils. Additionally (or alternatively), the multiple work coils **650A**, **650B** may represent different portions of a single overall drive coil, individual portions of which can be selectively energized, as discussed above in relation to FIG. 7A.

(59) If the multiple heater elements are being used to control the amount of vapor for a given puff, then having a larger number of heater elements, e.g. more than the two shown in FIG. 7A, will give a better granularity of control. It will also be appreciated that the vapor amount can be increased by supplying more electrical power to each work coil for energizing the corresponding heater element, however, there are limits on the practicality of this. For example, providing too much power may result in a very high temperature for the heater element, which may alter the chemical constitution of the vapor, as well as presenting a potential safety concern.

(60) FIG. 7B shows another implementation for supporting selectivity across multiple work coils **650A**, **650B**. Thus in FIG. 7B, it is assumed that the work coils **650A**, **650B** are not electrically connected to one another, but rather each work coil **650A**, **650B** is individually (separately) linked to the power source **782B** via a pair of independent connections through switch configuration **781B**. In particular, work coil **650A** is linked to power source **782B** via switch connections **A1** and **A2**, and work coil **650B** is linked to power source **782B** via switch connections **B1** and **B2**. This configuration of FIG. 7B offers similar advantages to those discussed above in relation to FIG. 7A. In addition, the architecture of FIG. 7B may also be readily scaled up to work with more than two work coils.

(61) FIG. 7C shows another implementation for supporting selectivity across multiple work coils, in this case three work coils denoted **650A**, **650B** and **650C**. Each work coil **650A**, **650B**, **650C** is directly connected to a respective power supply **782C1**, **782C2** and **782C3**. The configuration of FIG. 7 may support the selective energization of any single work coil, **650A**, **650B**, **650C**, or of any pair of work coils at the same time, or of all three work coils at the same time.

(62) In the configuration of FIG. 7C, at least some portions of the power supply **782** may be replicated for each of the different work coils **650**. For example, each power supply **782C1**, **782C2**, **782C3** may include its own inverter, but they may share a single, ultimate power source, such as battery **411**. In this case, the battery **411** may be connected to the inverters via a switch configuration analogous to that shown in FIG. 7B (but for Direct Current (DC) rather than AC current). Alternatively, each respective power line from a power supply **782** to a work coil **650** may

be provided with its own individual switch, which can be closed to activate the work coil (or opened to prevent such activation). In this arrangement, the collection of these individual switches across the different lines can be regarded as another form of switch configuration.

(63) There are various ways in which the switching of FIGS. 7A-7C may be managed or controlled. In some cases, the user may operate a mechanical or physical switch that directly sets the switch configuration. For example, e-cigarette **610** may include a switch (not shown in FIG. 6) on the outer housing, whereby cartridge **630A** can be activated in one setting, and cartridge **630B** can be activated in another setting. A further setting of the switch may allow activation of both cartridges together. Alternatively, the control unit **620** may have a separate button associated with each cartridge, and the user holds down the button for the desired cartridge (or potentially both buttons if both cartridges should be activated). Another possibility is that a button or other input device on the e-cigarette **610** may be used to select a stronger puff (and result in switching on both or all work coils). Such a button may also be used to select the addition of a flavor, and the switching might operate a work coil associated with that flavor—typically in addition to a work coil for the base liquid containing nicotine. The skilled person will be aware of other possible implementations of such switching.

(64) In some e-cigarettes, rather than direct (e.g. mechanical or physical) control of the switch configuration, the user may set the switch configuration via the comms/user interface **718** shown in FIG. 7 (or any other similar facility). For example, this comms/user interface **718** may allow a user to specify the use of different flavors or cartridges (and/or different strength levels), and the controller **715** can then set the switch configuration **781** according to this user input.

(65) A further possibility is that the switch configuration may be set automatically. For example, e-cigarette **610** may prevent work coil **650A** from being activated if a cartridge is not present in the illustrated location of cartridge **630A**. In other words, if no such cartridge is present, then the work coil **650A** may not be activated (thereby saving power, etc.).

(66) There are various mechanisms available for detecting whether or not a cartridge is present. For example, the control unit **620** may be provided with a switch which is mechanically operated by inserting a cartridge into the relevant position. If there is no cartridge in position, then the switch is set so that the corresponding work coil is not powered. Another approach would be for the control unit to have some optical or electrical facility for detecting whether or not a cartridge is inserted into a given position.

(67) Note that in some devices, once a cartridge has been detected as in position, then the corresponding work coil is always available for activation—e.g. it is always activated in response to a puff (inhalation) detection. In other devices that support both automatic and user-controlled switch configuration, even if a cartridge has been detected as in position, a user setting (or such-like, as discussed above) may then determine whether or not the cartridge is available for activation on any given puff.

(68) Although the control electronics of FIGS. 7A-7C have been described in connection with the use of multiple cartridges, such as shown in FIG. 6, they may also be utilized in respect of a single cartridge that has multiple heater elements. In other words, the control electronics is able to selectively energize one or more of these multiple heater elements within the single cartridge. Such an approach may still offer the benefits discussed above. For example, if the cartridge contains multiple heater elements, but just a single, shared reservoir, or multiple heater elements, each with its own respective reservoir, but all reservoirs containing the same liquid, then energizing more or fewer heater elements provides a way for a user to increase or decrease the amount of vapor provided with a single puff. Similarly, if a single cartridge contains multiple heater elements, each with its own respective reservoir containing a particular liquid, then energizing different heater elements (or combinations thereof) provides a way for a user to selectively consume vapors for different liquids (or combinations thereof).

(69) In some e-cigarettes, the various work coils and their respective heater elements (whether

implemented as separate work coils and/or heater elements, or as portions of a larger drive coil and/or susceptor) may all be substantially the same as one another, to provide a homogeneous configuration. Alternatively, a heterogeneous configuration may be utilized. For example, with reference to e-cigarette **610** as shown in FIG. **6**, one cartridge **630A** may be arranged to heat to a lower temperature than the other cartridge **630B**, and/or to provide a lower output of vapor (by providing less heating power). Thus if one cartridge **630A** contains the main liquid formulation containing nicotine, while the other cartridge **630B** contains a flavorant, it may be desirable for cartridge **630A** to output more vapor than cartridge **630B**. Also, the operating temperature of each heater element **655** may be arranged according to the liquid(s) to be vaporized. For example, the operating temperature should be high enough to vaporize the relevant liquid(s) of a particular cartridge, but typically not so high as to chemically break down (disassociate) such liquids.

(70) There are various ways of providing different operating characteristics (such as temperature) for different combinations of work coils and heater elements, and thereby produce a heterogeneous configuration as discussed above. For example, the physical parameters of the work coils and/or heater elements may be varied as appropriate—e.g. different sizes, geometry, materials, number of coil turns, etc. Additionally (or alternatively), the operating parameters of the work coils and/or heater elements may be varied, such as by having different AC frequencies and/or different supply currents for the work coils.

(71) In order to address various issues and advance the art, this disclosure shows by way of illustration various embodiments in which the claimed invention(s) may be practiced. The advantages and features of the disclosure are of a representative sample of embodiments only, and are not exhaustive and/or exclusive. They are presented only to assist in understanding and to teach the claimed invention(s). It is to be understood that advantages, embodiments, examples, functions, features, structures, and/or other aspects of the disclosure are not to be considered limitations on the disclosure as defined by the claims or limitations on equivalents to the claims, and that other embodiments may be utilized and modifications may be made without departing from the scope of the claims. Various embodiments may suitably comprise, consist of, or consist essentially of, various combinations of the disclosed elements, components, features, parts, steps, means, etc other than those specifically described herein. The disclosure may include other inventions not presently claimed, but which may be claimed in future.

Claims

1. An electronic vapor provision system having a longitudinal axis and comprising: a control unit including a control unit engagement end and a cartridge including a cartridge engagement end configured to engage with and disengage from one another substantially along said longitudinal axis, and the control unit including a battery and a tube portion disposed along said longitudinal axis, wherein the tube portion comprises an inner wall and an outer wall; and an induction heating assembly comprising a drive coil and a heater element, wherein the heater element is located in the cartridge, the drive coil being located in the tube portion of the control unit, between the inner wall and the outer wall of the tube portion, and disposed about said longitudinal axis, wherein the cartridge is located at least partly within the inner wall of the tube portion when engaged with said control unit, such that the heater element is within the drive coil, wherein the control unit is configured such that when the cartridge is engaged with the control unit, a gap is formed between the cartridge engagement end and the control unit engagement end, whereby, in use, air is able to be drawn from the gap through an air inlet of the cartridge provided in the cartridge engagement end; and wherein the control unit is provided with one or more holes in a housing of the control unit to allow, in use, air to be drawn into the control unit and into the gap.
2. The electronic vapor provision system of claim 1, wherein the cartridge comprises a mouthpiece.
3. The electronic vapor provision system of claim 1, wherein the heater element is substantially

planar and substantially perpendicular to the longitudinal axis.

4. The electronic vapor provision system of claim 1, wherein the heater element is made of a material having a steel.

5. The electronic vapor provision system of claim 1, wherein the heater element is made of a material having a ferrous steel.

6. The electronic vapor provision system of claim 1, wherein there is no wired electrical connection between the control unit and the cartridge.

7. The electronic vapor provision system of claim 1, wherein the cartridge does not have any external metal connector for electrical connection between the cartridge and the control unit.

8. The electronic vapor provision system of claim 1, wherein the control unit accommodates multiple cartridges.

9. A control unit for use in an electronic vapor provision system having a longitudinal axis, the control unit including a control unit engagement end being configured to engage with and disengage from a cartridge including a cartridge engagement end substantially along said longitudinal axis, the control unit comprising: a battery and a tube portion disposed along said longitudinal axis, wherein the tube portion comprises an inner wall and an outer wall; and a drive coil being located in the tube portion of the control unit, between the inner wall and the outer wall of the tube portion, and disposed about said longitudinal axis; the drive coil being configured to form an induction heating assembly with a heater element located in the cartridge, wherein the cartridge is located at least partly within the inner wall of the tube portion when engaged with said control unit, such that the heater element is within the drive coil, wherein the control unit is configured such that when the cartridge is engaged with the control unit, a gap is formed between the cartridge engagement end and the control unit engagement end, whereby, in use, air is able to be drawn from the gap through an air inlet of the cartridge provided in the cartridge engagement end; and wherein the control unit is provided with one or more holes in a housing of the control unit to allow, in use, air to be drawn into the control unit and into the gap.

10. A cartridge for use in an electronic vapor provision system having a longitudinal axis, the cartridge being configured to engage with and disengage from a control unit substantially along said longitudinal axis, the control unit including a control engagement end and the cartridge including a cartridge engagement end, the control unit including a battery and a tube portion disposed along said longitudinal axis, wherein the tube portion comprises an inner wall and an outer wall, the cartridge comprising: a heater element located in the cartridge, the heater element being configured to form an induction heating assembly with a drive coil located in the tube portion of the control unit, between the inner wall and the outer wall of the tube portion, and disposed about said longitudinal axis, wherein the cartridge is located at least partly within the inner wall of the tube portion when engaged with said control unit, such that the heater element is within the drive coil, wherein the control unit is configured such that when the cartridge is engaged with the control unit, a gap is formed between the cartridge engagement end and the control unit engagement end, whereby, in use, air is able to be drawn from the gap through an air inlet of the cartridge provided in the cartridge engagement end; wherein the control unit is provided with one or more holes in a housing of the control unit to allow, in use, air to be drawn into the control unit and into the gap.
